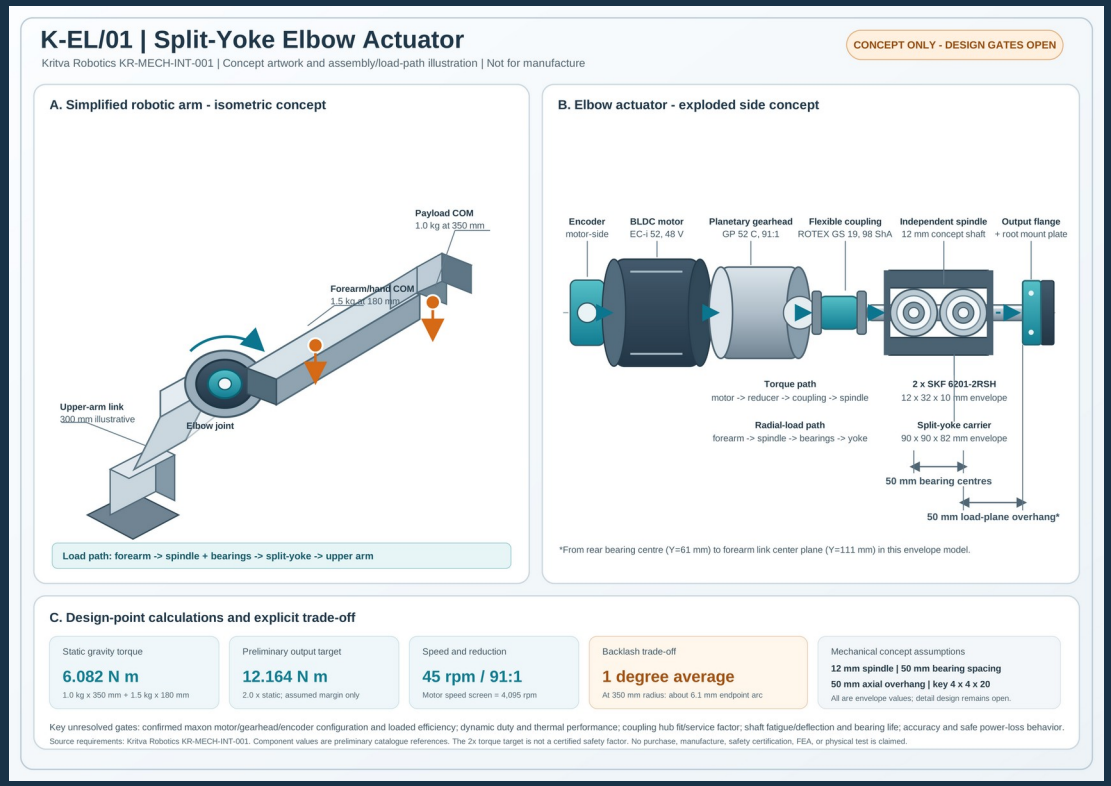
K-EL/01 Split-Yoke Elbow Actuator

Conceptual mechanical design • KR-MECH-INT-001

10-minute design walkthrough
Presenter: Candidate
25 October 2026



1. Problem & design basis

Use the brief inputs without silently inventing a full mission profile

Payload: 1.0 kg at 350 mm from elbow axis

Forearm + hand: 1.5 kg at 180 mm

Horizontal forearm is the initial gravity-only case

Output target 45 RPM; BLDC/PMSM + encoder; planetary preferred

Illustrative link envelope: 300 mm upper arm / 350 mm forearm

No duty cycle, acceleration, position accuracy, impact or power-loss holding requirement was supplied.

2. Static torque & preliminary target

Gravity-only result is a baseline, not a complete motion design

Term	Formula	Result
Payload	1.0 × 9.81 × 0.350	3.4335 N·m
Forearm/hand	1.5 × 9.81 × 0.180	2.6487 N·m
Static total	3.4335 + 2.6487	6.0822 N·m
Preliminary target	2.0 × static	12.1644 N·m
Ideal output power	12.1644 × 45×2π/60	57.32 W

The 2× multiplier is an assumption, not a certified safety factor.

Dynamics and inertia can dominate peak/RMS sizing.

3. Drive candidate & key caution

A shortlist is not a frozen supplier configuration

Motor candidate: maxon EC-i 52, 48 V, part 579165; nominal 4470 RPM, 0.423 N·m, 4.59 A.

Gearhead candidate: maxon GP 52 C, 91:1, part 223094; nominal input speed required = 4095 RPM.

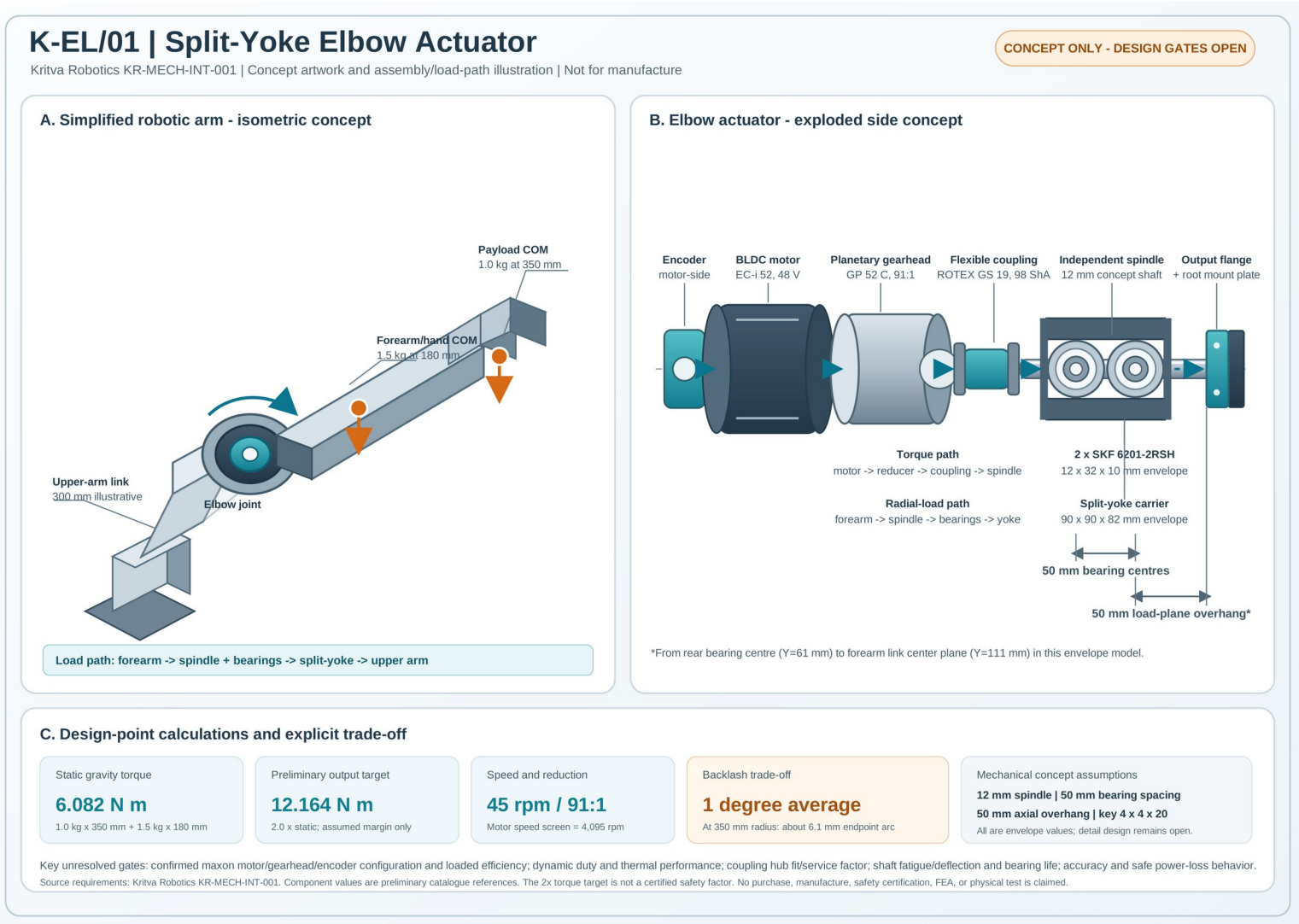
At assumed 40% gearhead efficiency: ~0.334 N·m motor torque, ~143 W input screen, ~3.64 A torque-constant current screen.

At assumed 30% efficiency, simple continuous torque/current screen fails.

Product title says 180 W class; published nominal T×ω is about 198 W. Ask maxon for authoritative interpretation and loaded curves.

4. Mechanical load path

Torque transmission and radial support are deliberately separated



Torque: motor → GP52C → ROTEX GS coupling → independent spindle → flange/root plate → forearm.

Radial load: forearm → spindle → two 6201 bearings → split yoke → upper arm.

Bearing centres 50 mm apart; forearm load plane is 50 mm beyond rear-bearing centre.

STEP is simplified envelope CAD: 21 solids, valid geometry; not proof of no interference or strength.

5. Structural & accuracy screens

Useful for concept decisions; not final verification

Gravity force from specified masses: 24.525 N; overhang moment: 1.22625 N·m = 1226.25 N·mm.

Ø12 mm nominal spindle: bending 7.23 MPa; torsional shear 35.85 MPa; von Mises ~62.52 MPa.

Stress screens omit keyway/notch factors, fatigue, shock, dynamic loads, detailed spindle geometry and deflection.

At 1° average no-load backlash, payload-radius arc is ~6.11 mm at 350 mm.

The final position error also includes compliance, controller/encoder, assembly and load effects.

6. Major risks & next evidence

Close mission requirements and vendor evidence before freezing hardware

Thermal/duty risk: define trajectory, acceleration, cycles and ambient; obtain loaded motor curve and gearhead efficiency; calculate peak/RMS torque.

Mechanical risk: confirm coupling hubs/keying, spindle section/fits, bearing life, carrier stiffness and fasteners.

Safety risk: clarify gravity hold during power loss; add rated normally-engaged brake or explicitly limit to externally supported supervised use.

Accuracy risk: 1° no-load backlash may be too high; define output-position tolerance and evaluate lower-backlash reducer/output feedback.

Validate with a guarded supervised test only after the configuration is confirmed.

7. Conclusion & status

Concept submission complete; engineering release is intentionally not claimed

Static target: 6.0822 N·m required; 12.1644 N·m preliminary target.

Candidate 91:1 drive is kinematically plausible but needs supplier-confirmed loaded operation and thermal validation.

Included package: report, workbook/CSVs, BOM, official reference matrix, STEP/source/macro, concept artwork, QA/gap plan, verification plan, presentation and discussion notes.

Native .FCStd was not created because FreeCAD is unavailable in this environment; run included macro on FreeCAD to create it.

No purchase, manufacture, FEA, physical test or certification is claimed.